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Third International Conference on Networking and Communications : Submission (2236) has been created.

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Microsoft CMT <noreply@msr-cmt.org> 22 February 2026 at 23:07
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The following submission has been created.

Track Name: ICNWC2026

Paper ID: 2236

Paper Title: Agri-Neuro: A Context-Aware Plant Disease Diagnosis Framework utilizing Hybrid Deep Learning and IoT-Driven Environmental Sensing

Abstract:
Plant diseases represent a critical threat to global food security, often necessitating early intervention to prevent catastrophic yield losses. Conventional diagnostic methods, relying heavily on manual expert inspection or standalone computer vision systems, are plagued by subjectivity, high latency, and an inability to distinguish between visually similar pathologies that thrive in disparate environments. This paper proposes AgriNeuro, a novel, integrated framework that synergizes Deep Learning (DL) with the Internet of Things (IoT) for realtime, precision disease diagnosis. Unlike existing solutions that treat visual and environmental data in isolation, our system employs a unique Context-Aware Weighted Fusion Algorithm. This mechanism correlates visual symptoms detected via a finetuned ResNet-50 Convolutional Neural Network (CNN) with real-time environmental telemetry (soil moisture, temperature, humidity) to calculate a comprehensive Disease Risk Probabilistic Score (DRPS). Evaluated on a hybrid dataset comprising 54,306 images and 500 field-verified samples, our fusion-based approach achieves a classification accuracy of 96.4%, outperforming standalone EfficientNet-B0 and ResNet-50 models by 2.1% and 4.2% respectively. Furthermore, the system successfully predicts latent infection risks in 89% of test cases where visual symptoms were non-manifest, proving the efficacy of integrating environmental context into disease diagnosis. The proposed architecture offers a scalable, low-cost solution for precision agriculture in resourceconstrained settings.

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Submission Questions Response: Not Entered

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